

LittlePath Homeschool

Final MVP Product + Developer Specification

16 Locked Flows • Desktop + Mobile UI References • July 2026

Status	Locked MVP specification for implementation
Roles	Admin / Head of School, Teacher, Parent
Initial scale	Fewer than 10 teachers
AI budget	Target approximately US\$30/month initially
Approved AI	Teacher voice-note processing and intelligent scheduling
UI source	16 embedded UI boards plus this written functional specification

1. Product Definition

LittlePath is a lightweight homeschool learning portal for curriculum, students, teachers, scheduling, attendance, class records, academic learning progress, development observations, media moments and parent weekly updates. The MVP is deliberately focused on reducing administrative work without becoming a large school ERP.

Locked product principles

- Keep Home calm. Detailed functionality belongs in its dedicated tab rather than being duplicated on Home.
- Learning = academic curriculum progress. Development = personal, emotional and physical observations. Moments = photos/videos with lightweight context. Keep these separate.
- Teachers document a class once. A single natural voice recording must be split into an overall Class Note plus individual student observations.
- Always preserve the overall Class Note, even when no student-specific optional update is made.
- AI is used only where it creates material workflow savings; it is not required for ordinary dashboards or structured-data features.
- Admin/teacher review precedes persistence of AI-generated or AI-proposed changes.

2. Roles & Permissions

Admin	Full school workspace: curriculum, teachers, students, schedules, operational issues, records, Learning, Development and Moments.
Teacher	Own/relevant classes, attendance, class records, student observations and permitted Learning/Development/Moment updates.
Parent	Linked child only: Home, Learning, Development, Moments and weekly update. No Schedule and no Ask Teacher in MVP.

3. Locked MVP Boundaries

- No parent Schedule view.
- No Ask Teacher or parent-teacher messaging anywhere.

- No global search.
- No library integration.
- No OTP login; school-issued username and password.
- Parents receive only one push notification per week: weekly update ready.
- Teachers receive notifications only for scheduling and class-update/class-record needs.
- Attendance is part of the class-record/post-class workflow.
- Moments is media-focused; Development remains separate.
- Advanced analytics/reporting, homework/assignments and extra integrations are deferred.

4. Core Data Model

User	username, password hash, role, status, profile
Teacher	user, teaching capabilities, regular availability, temporary exceptions
Parent	user, linked student(s)
Student	profile, status, parent access, subject/level assignments, availability
Curriculum	Subject → Level → Topic → Skill
Class	subject/level, teacher, students, recurrence, duration, scheduled time
Class Occurrence	specific scheduled instance; anchor for attendance and class record
Attendance	occurrence + student + Present/Absent/Late
Class Record	occurrence + overall Class Note + author + timestamps
Student Observation	class record + student + observation
Learning Update	student + skill + status + note/evidence + source
Development Observation	student + area + observation + evidence + source
Moment	media + title/caption + context + tagged students
Weekly Update	student + week + selected Learning/Development/Moments/class-note content
Notification	recipient + type + state + timestamps

Critical relationship: the Class Occurrence is the common anchor for attendance, overall Class Note, individual student observations, optional Learning updates, optional Development updates and Moments. Do not create duplicate class histories.

5. AI Architecture & Budget

- Target approximately US\$30/month initially.
- Voice workflow: speech-to-text → language handling → identify overall class content → identify attending/named children → split child-specific observations → produce editable structured draft.
- The teacher may speak naturally about the whole class and individual children in one recording. The system must produce BOTH the overall Class Note and individual student observations.
- Scheduling workflow: AI interprets natural-language intent; deterministic application logic validates teacher capability, teacher availability, student availability, existing classes and conflicts.
- AI must not autonomously mutate schedules or class records before explicit review/confirmation.
- Do not spend AI on routine dashboards, Learning calculations, Development stages, Moments, notifications or authentication.
- Track AI usage/cost by workflow.

6. Final Flow Index

01. Admin Home / Needs Attention — Admin: Operational home for today's classes and actionable issues.
02. Parent Weekly Update + Notifications + Settings — Parent: Weekly parent update experience and basic settings.
03. Curriculum Management — Admin: Create subjects, levels, topics and skills.
04. Setup, Empty, Loading, Error & Offline States — All relevant roles: First-use setup and system states.
05. Admin & Teacher Home States — Admin / Teacher: Daily operational home variants.
06. AI-Assisted Class Scheduling — Admin: Create recurring classes from natural language.
07. Class Record — Attendance, Voice Note & AI Review — Teacher: Create the post-class record from one voice note.
08. Authentication & Basic Settings — All roles: Username/password access and settings.
09. Student Management — Admin: Create and maintain students.
10. Teacher Management — Admin: Manage fewer than 10 teachers, capabilities and availability.
11. Parent Portal — Home, Learning, Development & Moments — Parent: Low-clutter view of the child's learning and growth.
12. Moments — Admin / Teacher: Media gallery and add/tag workflow.
13. Development — Admin / Teacher: Evidence-based development observations.
14. Student Learning Map & Skill Progress — Admin / Teacher: Track academic curriculum progress per student.
15. Schedule Management & Voice Rescheduling — Admin: View schedule and propose multi-class changes.
16. Teacher Post-Class Workflow — Teacher: Teacher-first end-to-end class documentation flow.

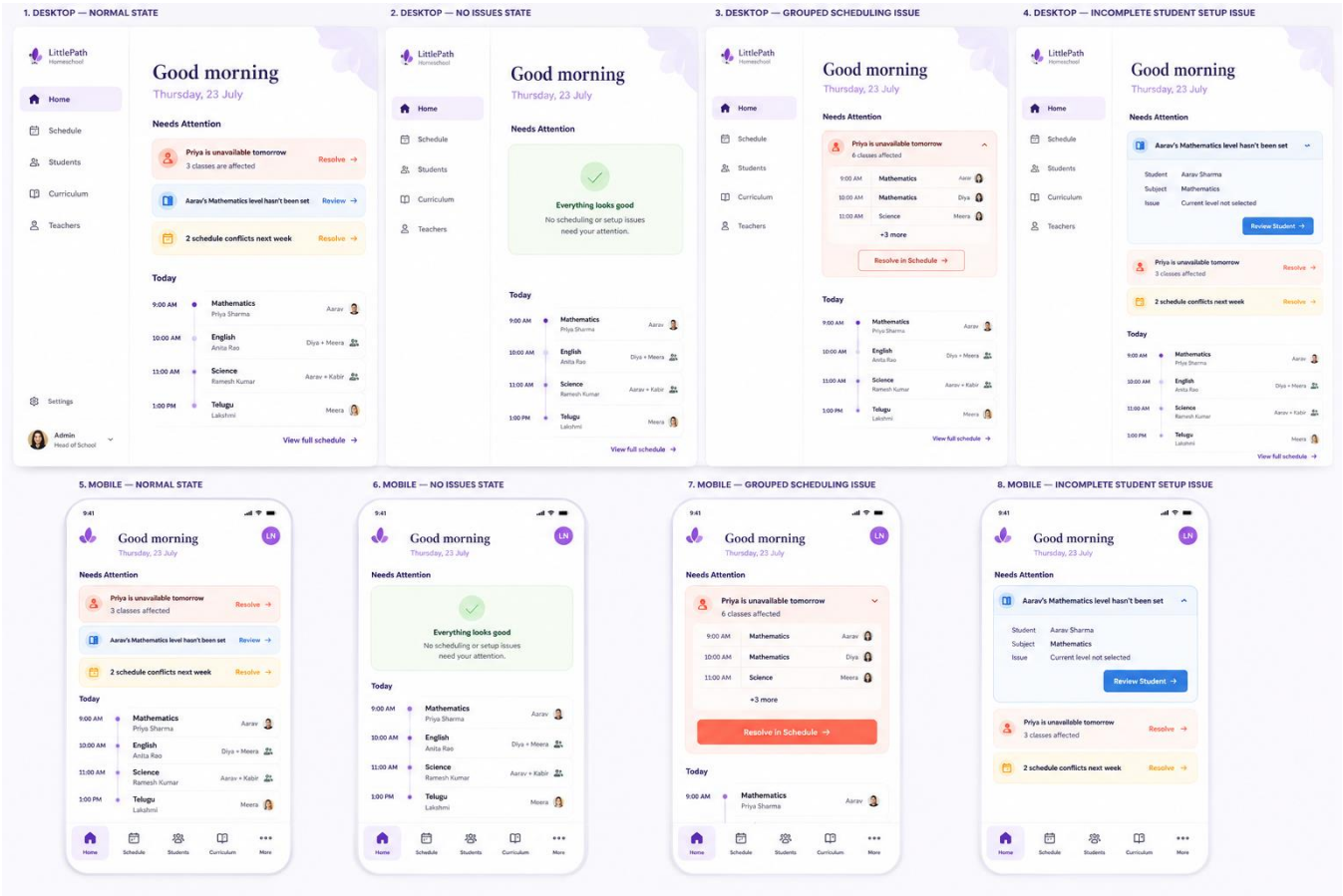
Flow 01 — Admin Home / Needs Attention

Primary role	Admin
Purpose	Operational home for today's classes and actionable issues.

Locked behavior

- Home is an operational summary, not a duplicate of Schedule/Students/Curriculum.
- Show greeting/date, Needs Attention and today's classes. No-issue state should be calm.
- Group related scheduling issues and allow drill-down to affected classes.
- Incomplete student setup routes to the relevant student record. No search or dashboard analytics.

Official desktop + mobile UI reference



Flow 01 — Admin Home / Needs Attention

Source-of-truth rule for this flow: the written specification controls behavior, permissions and data meaning; this board controls intended visual structure, responsive layout and screen states. If generated sample names/dates/copy conflict with a locked MVP rule, the written rule takes precedence.

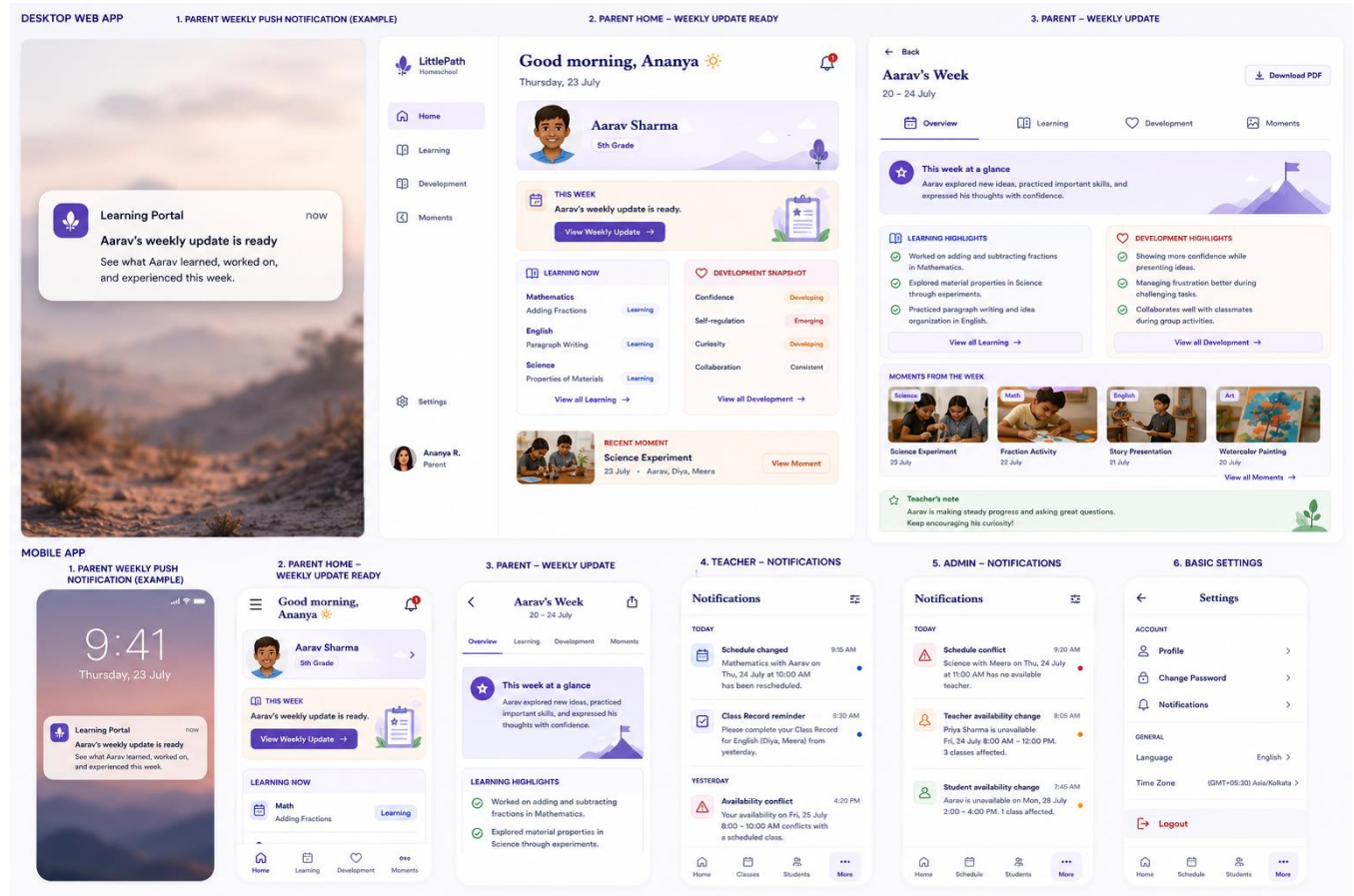
Flow 02 — Parent Weekly Update + Notifications + Settings

Primary role	Parent
Purpose	Weekly parent update experience and basic settings.

Locked behavior

- One weekly push only: weekly update ready.
- Home contains child card, weekly-update CTA, compact Learning Now, Development Snapshot and recent Moment.
- Weekly Update separates Overview, Learning, Development and Moments.
- Parent has no Schedule and no Ask Teacher.

Official desktop + mobile UI reference



Flow 02 — Parent Weekly Update + Notifications + Settings

Source-of-truth rule for this flow: the written specification controls behavior, permissions and data meaning; this board controls intended visual structure, responsive layout and screen states. If generated sample names/dates/copy conflict with a locked MVP rule, the written rule takes precedence.

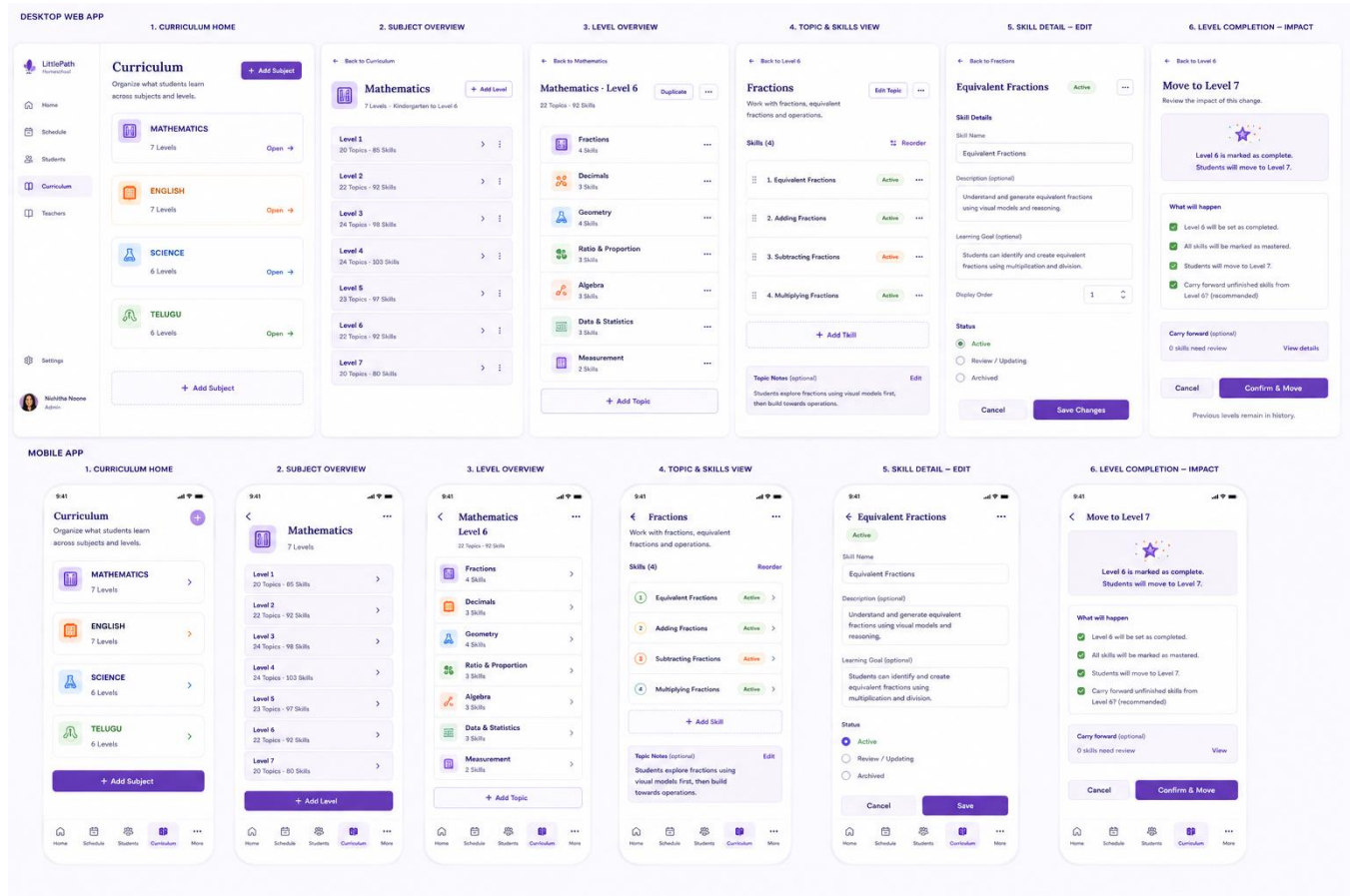
Flow 03 — Curriculum Management

Primary role	Admin
Purpose	Create subjects, levels, topics and skills.

Locked behavior

- Hierarchy: Subject → Level → Topic → Skill.
- Admin can create/edit structure, learning goals, ordering and active/archive state.
- Level completion previews impact and supports explicit confirmation/carry-forward handling.
- Curriculum definitions are separate from student-specific mastery.

Official desktop + mobile UI reference



Flow 03 — Curriculum Management

Source-of-truth rule for this flow: the written specification controls behavior, permissions and data meaning; this board controls intended visual structure, responsive layout and screen states. If generated sample names/dates/copy conflict with a locked MVP rule, the written rule takes precedence.

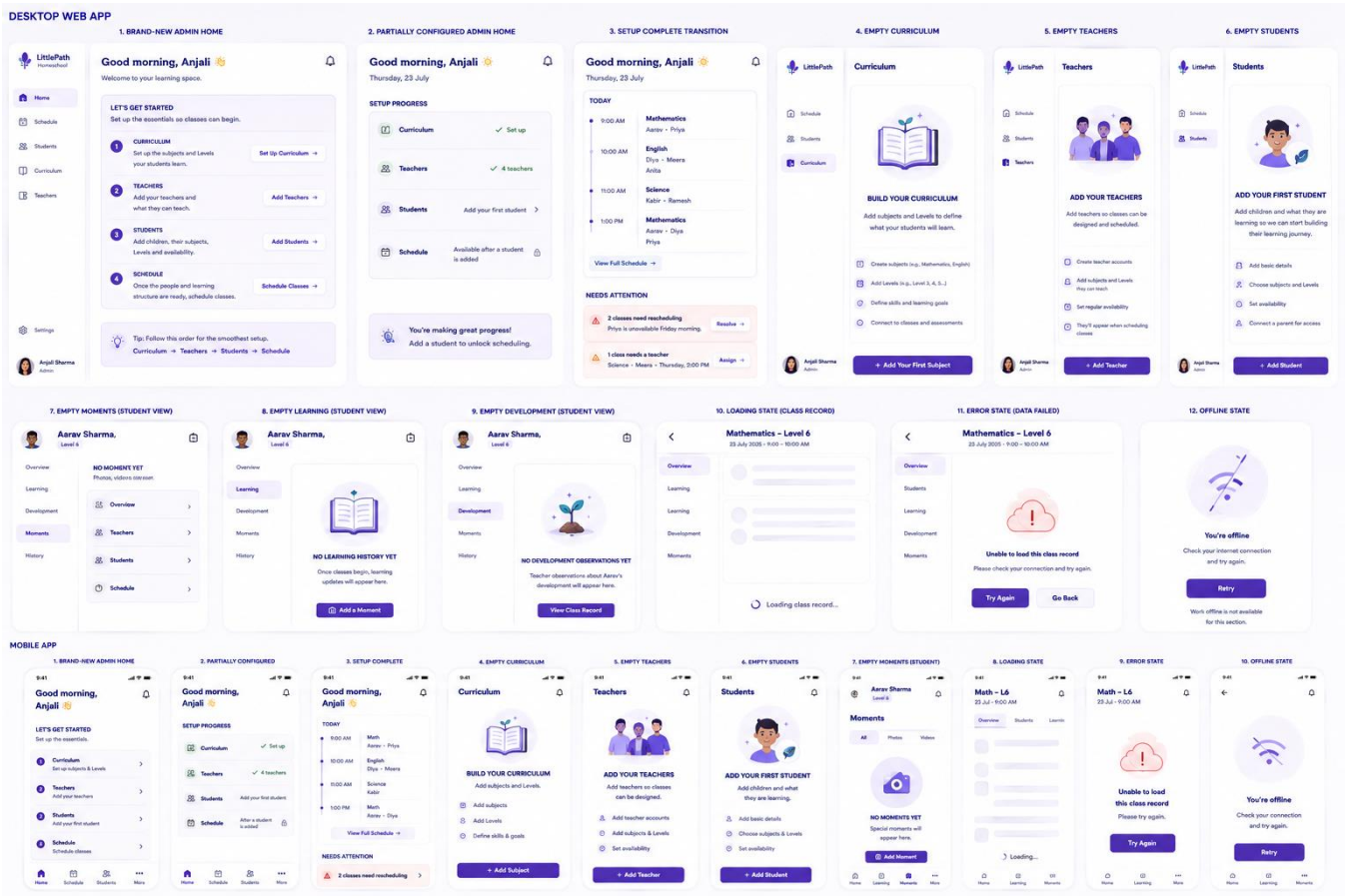
Flow 04 — Setup, Empty, Loading, Error & Offline States

Primary role	All relevant roles
Purpose	First-use setup and system states.

Locked behavior

- Initial admin sequence: Curriculum → Teachers → Students → Schedule.
- Partially configured Home shows setup progress and next action.
- Provide empty states for Curriculum, Teachers, Students, Moments, Learning and Development.
- Explicit loading, failed-data and offline states; do not imply unsupported offline editing.

Official desktop + mobile UI reference



Flow 04 — Setup, Empty, Loading, Error & Offline States

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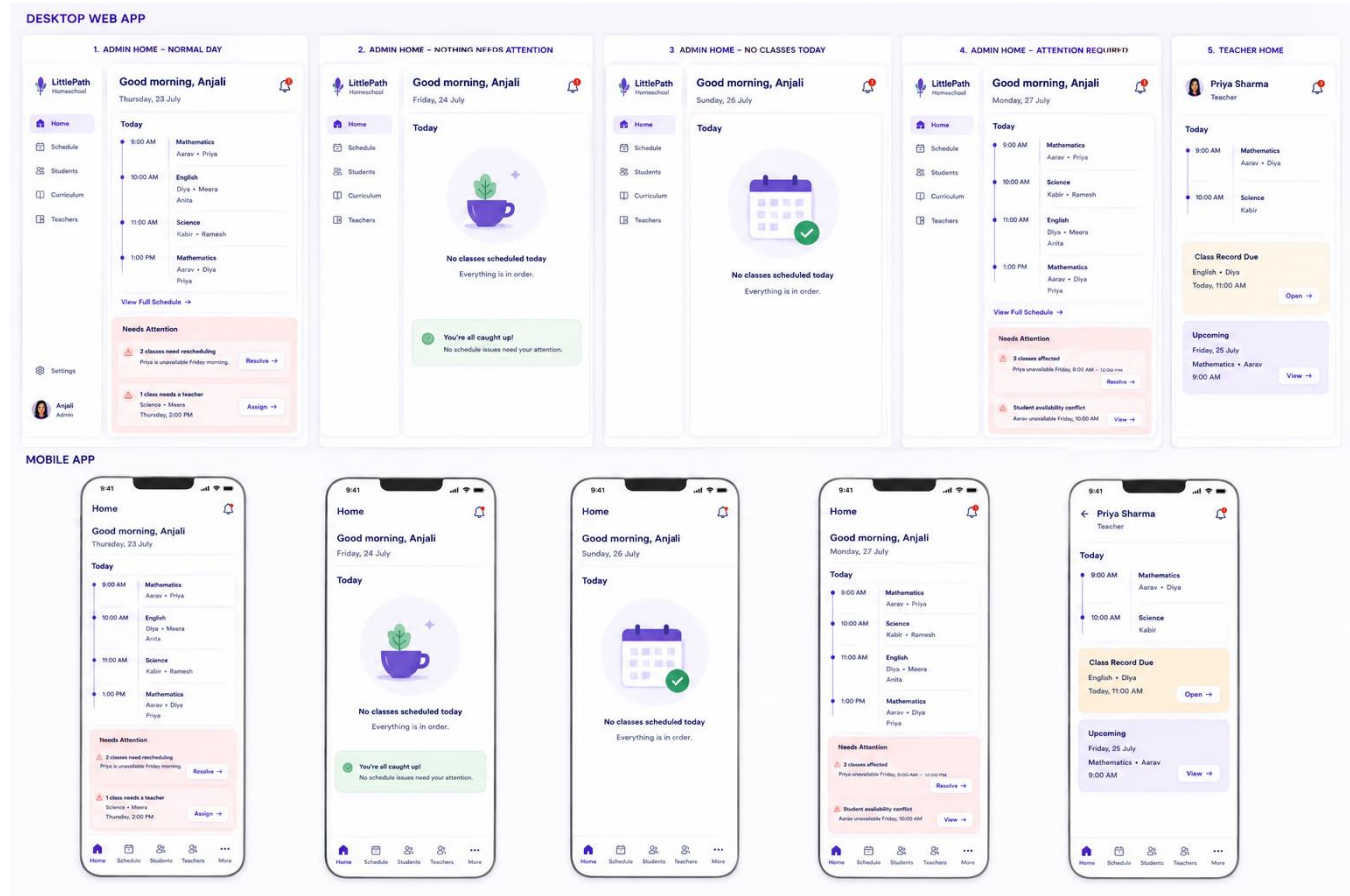
Flow 05 — Admin & Teacher Home States

Primary role	Admin / Teacher
Purpose	Daily operational home variants.

Locked behavior

- Admin Home prioritizes today's classes and Needs Attention.
- No classes today is a valid calm state.
- Attendance/class-record-required states route to the class.
- Teacher Home shows only relevant classes, class records due and upcoming classes.

Official desktop + mobile UI reference



Flow 05 — Admin & Teacher Home States

Source-of-truth rule for this flow: the written specification controls behavior, permissions and data meaning; this board controls intended visual structure, responsive layout and screen states. If generated sample names/dates/copy conflict with a locked MVP rule, the written rule takes precedence.

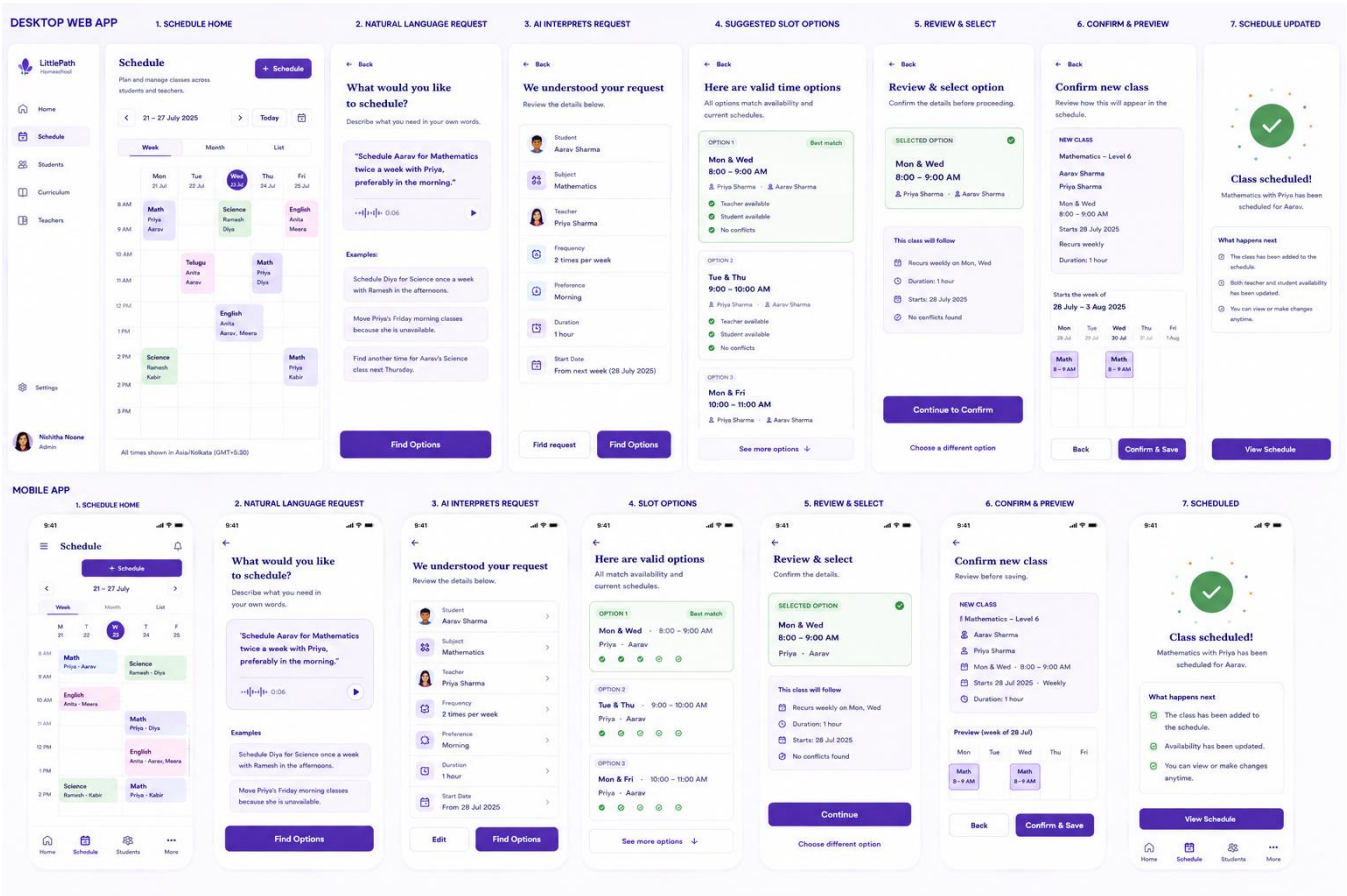
Flow 06 — AI-Assisted Class Scheduling

Primary role	Admin
Purpose	Create recurring classes from natural language.

Locked behavior

- Admin enters/speaks student, subject, teacher, frequency, time preference, duration and start-date intent.
- AI converts request into reviewable structured fields.
- Constraint engine returns only valid options based on capabilities, teacher/student availability, existing schedule and conflicts.
- Admin selects, reviews and confirms before schedule is written.

Official desktop + mobile UI reference



Flow 06 — AI-Assisted Class Scheduling

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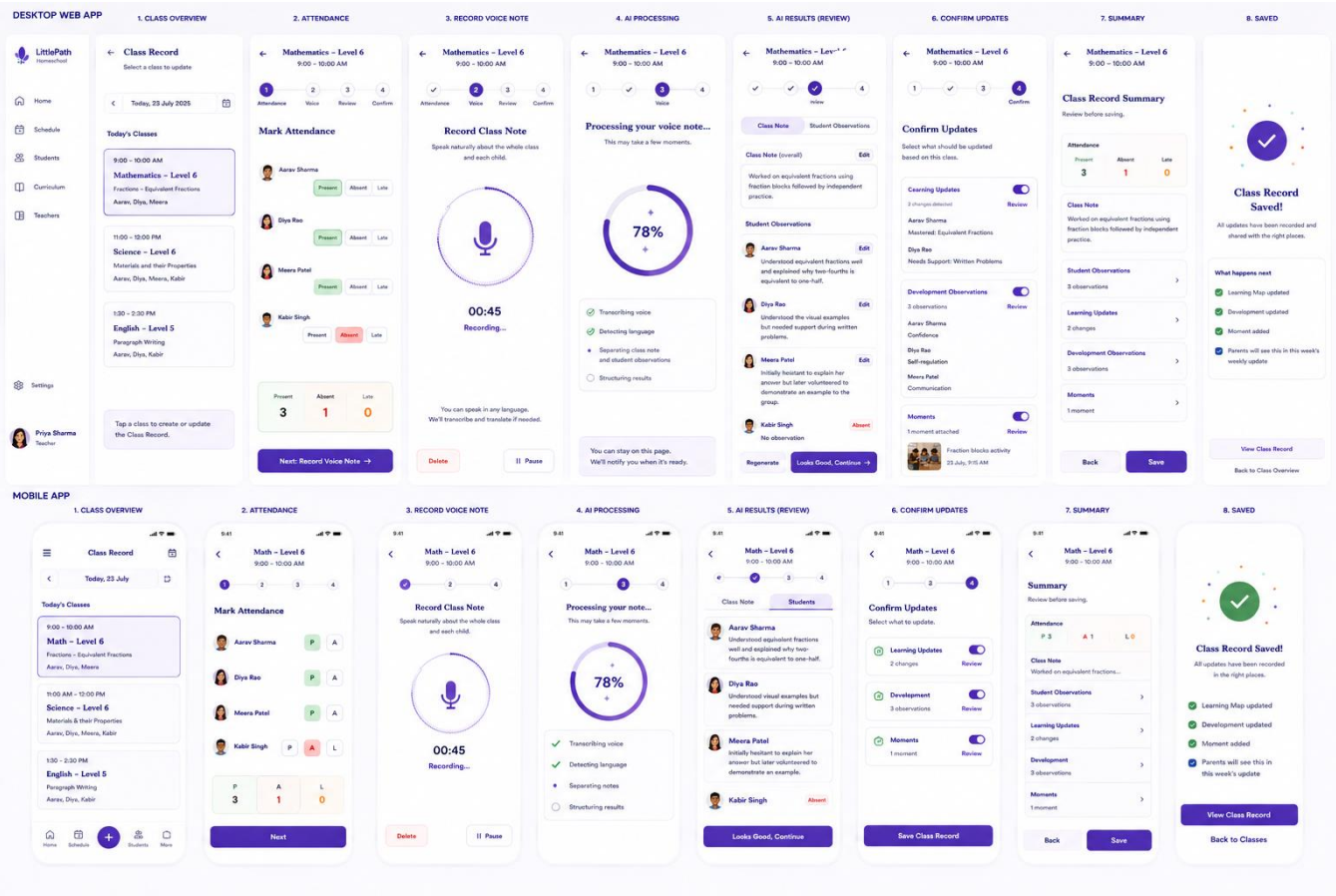
Flow 07 — Class Record — Attendance, Voice Note & AI Review

Primary role	Teacher
Purpose	Create the post-class record from one voice note.

Locked behavior

- Select occurrence → attendance → one natural voice note → AI processing → review → confirm optional updates → summary → save.
- AI Results separates Overall Class Note from Student Observations and both are editable.
- Learning, Development and Moment updates are optional.
- Overall Class Note is always preserved.
- Prefer an atomic multi-part save or a recoverable failure state.

Official desktop + mobile UI reference



Flow 07 — Class Record — Attendance, Voice Note & AI Review

Source-of-truth rule for this flow: the written specification controls behavior, permissions and data meaning; this board controls intended visual structure, responsive layout and screen states. If generated sample names/dates/copy conflict with a locked MVP rule, the written rule takes precedence.

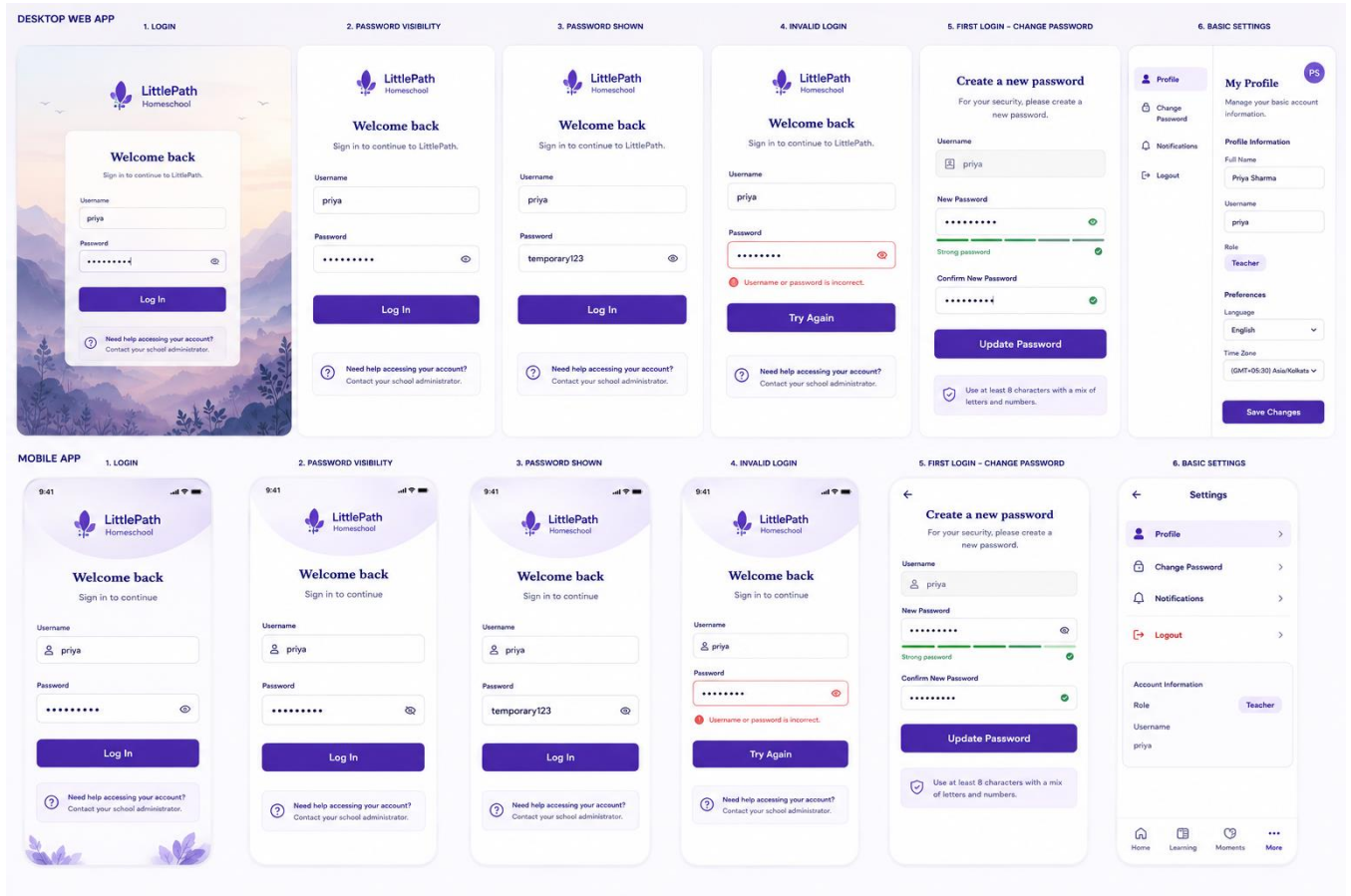
Flow 08 — Authentication & Basic Settings

Primary role	All roles
Purpose	Username/password access and settings.

Locked behavior

- School provides username and temporary password; no OTP.
- Support password visibility and generic invalid-login error.
- First login forces password replacement.
- Role is not user-editable; settings remain basic.

Official desktop + mobile UI reference



Flow 08 — Authentication & Basic Settings

Source-of-truth rule for this flow: the written specification controls behavior, permissions and data meaning; this board controls intended visual structure, responsive layout and screen states. If generated sample names/dates/copy conflict with a locked MVP rule, the written rule takes precedence.

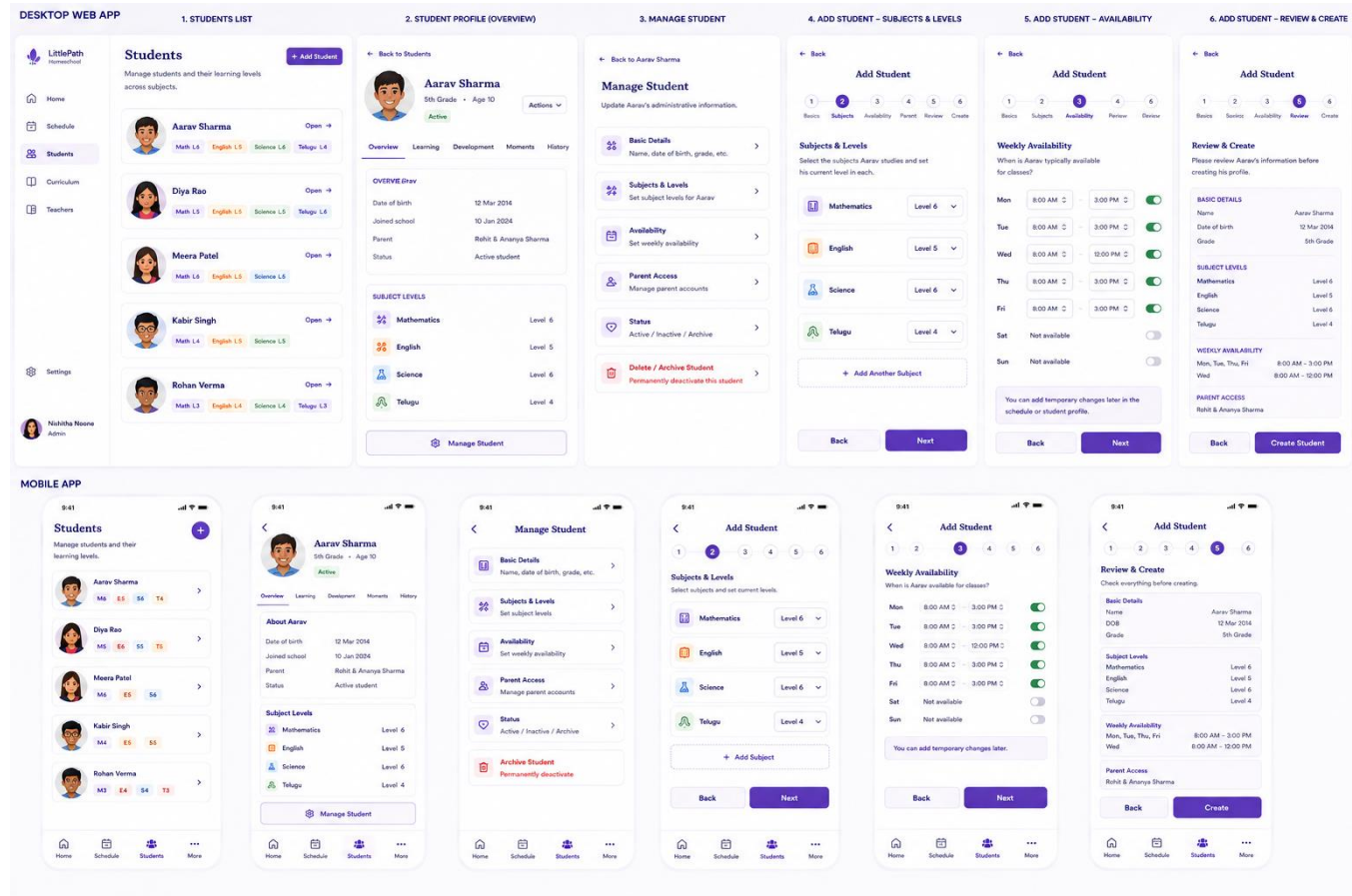
Flow 09 — Student Management

Primary role	Admin
Purpose	Create and maintain students.

Locked behavior

- Student profile connects Overview, Learning, Development, Moments and relevant history.
- Manage Basic Details, Subjects & Levels, Availability, Parent Access and Status.
- Add Student captures current subject levels, weekly availability and parent access before review/create.
- Archive/deactivate rather than destructively deleting historical records.

Official desktop + mobile UI reference



Flow 09 — Student Management

Source-of-truth rule for this flow: the written specification controls behavior, permissions and data meaning; this board controls intended visual structure, responsive layout and screen states. If generated sample names/dates/copy conflict with a locked MVP rule, the written rule takes precedence.

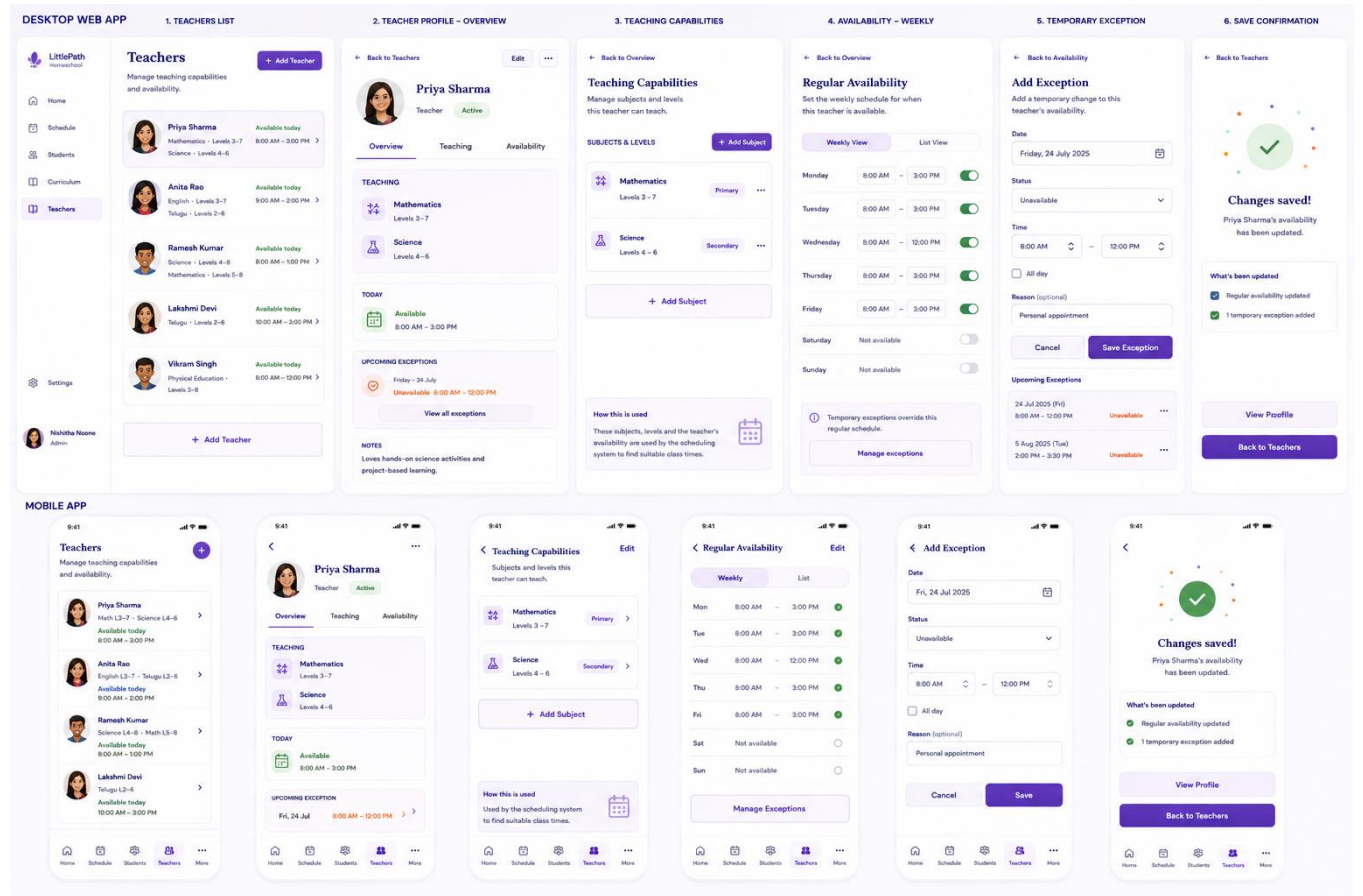
Flow 10 — Teacher Management

Primary role	Admin
Purpose	Manage fewer than 10 teachers, capabilities and availability.

Locked behavior

- Keep teacher list simple for the initial small team.
- Teaching Capabilities maps subject and level ranges.
- Regular Availability is weekly recurring availability.
- Temporary Exceptions override regular availability on specific dates/times.
- Changes feed schedule validation/conflict detection.

Official desktop + mobile UI reference



Flow 10 — Teacher Management

Source-of-truth rule for this flow: the written specification controls behavior, permissions and data meaning; this board controls intended visual structure, responsive layout and screen states. If generated sample names/dates/copy conflict with a locked MVP rule, the written rule takes precedence.

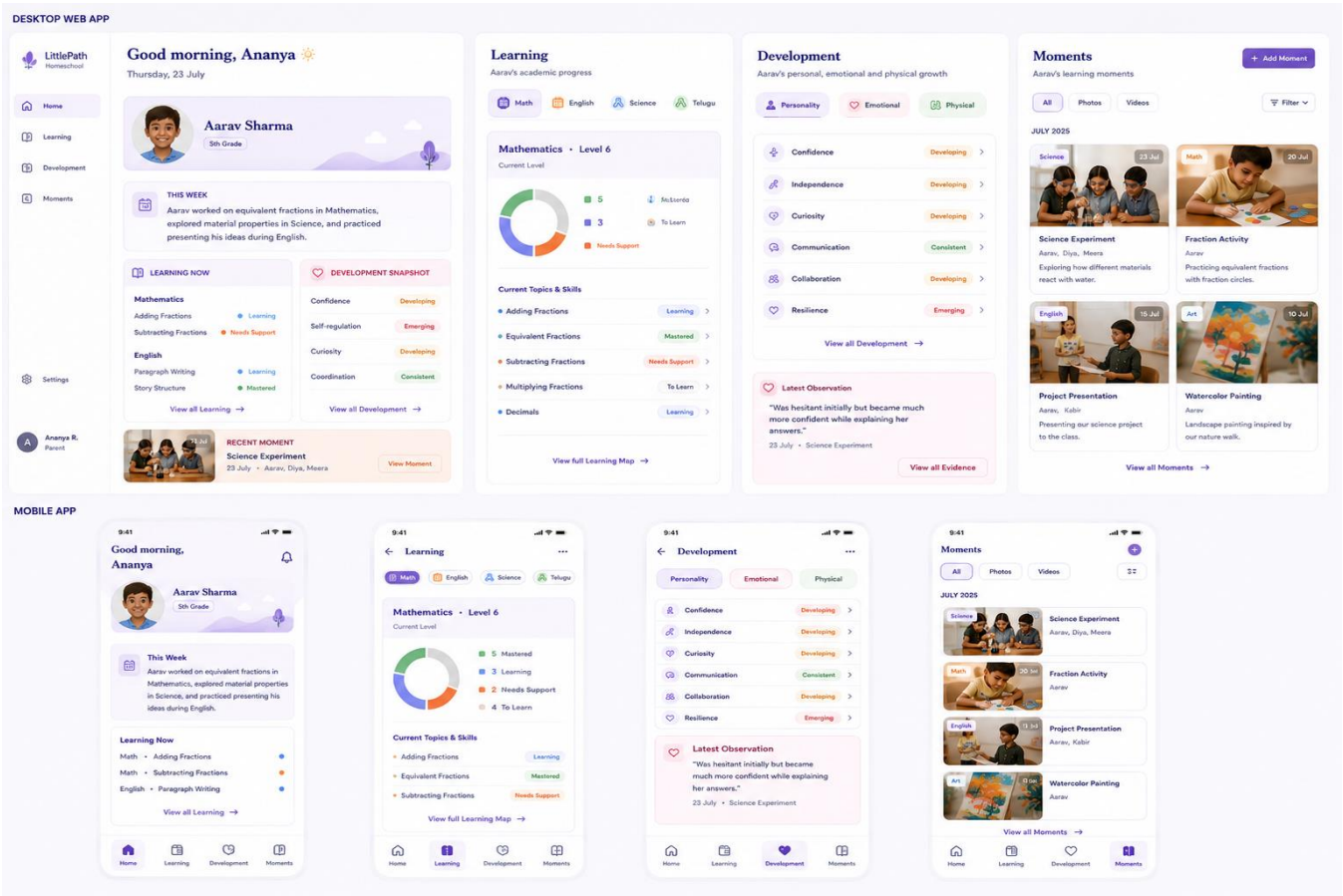
Flow 11 — Parent Portal — Home, Learning, Development & Moments

Primary role	Parent
Purpose	Low-clutter view of the child's learning and growth.

Locked behavior

- Navigation: Home, Learning, Development, Moments.
- Home summarizes; tabs hold detail.
- Learning is academic skill progress; Development is evidence-based growth; Moments is media.
- No Ask Teacher and no Schedule.

Official desktop + mobile UI reference



Flow 11 — Parent Portal — Home, Learning, Development & Moments

Source-of-truth rule for this flow: the written specification controls behavior, permissions and data meaning; this board controls intended visual structure, responsive layout and screen states. If generated sample names/dates/copy conflict with a locked MVP rule, the written rule takes precedence.

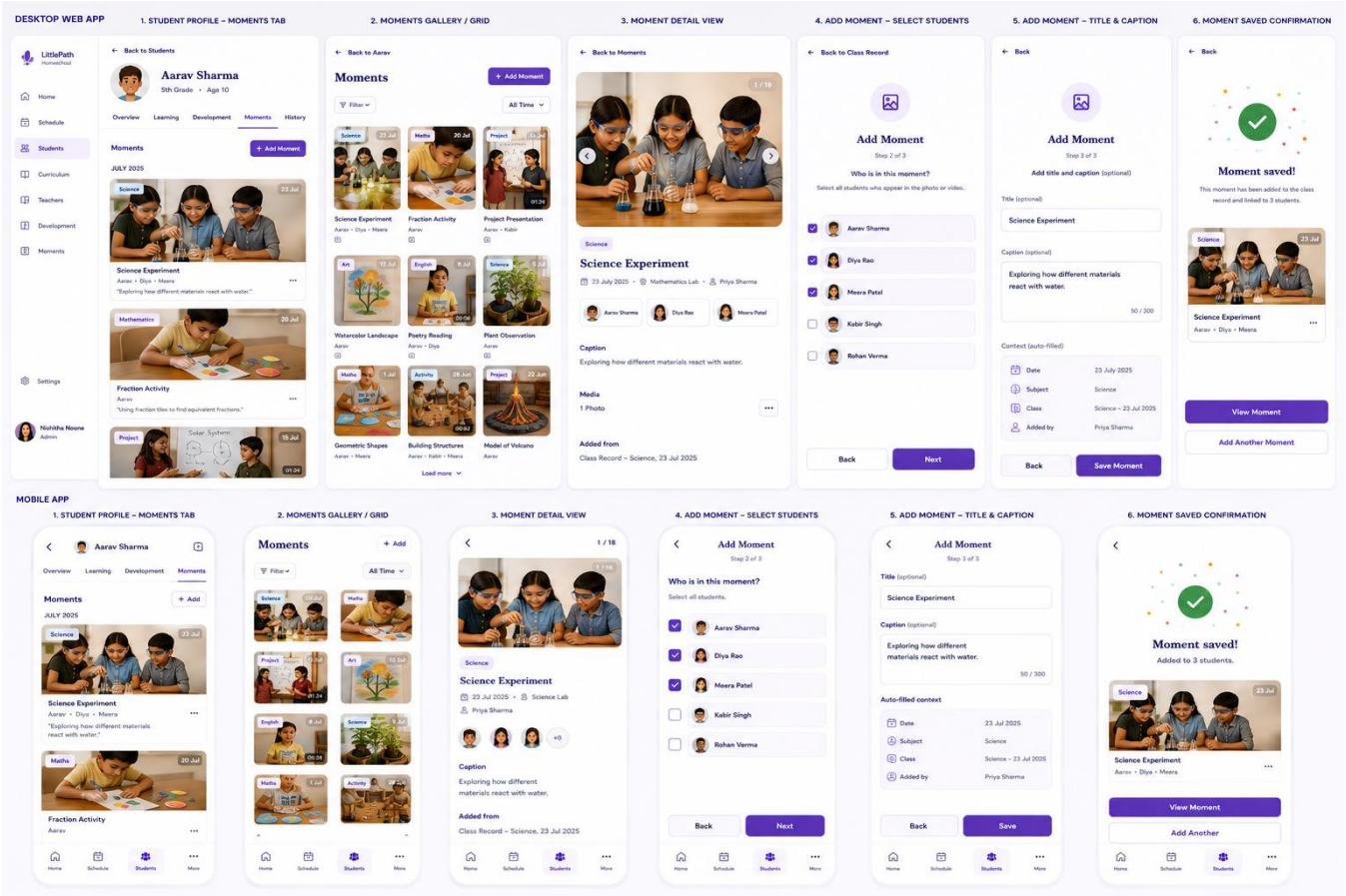
Flow 12 — Moments

Primary role	Admin / Teacher
Purpose	Media gallery and add/tag workflow.

Locked behavior

- Moment detail stores media, tagged students, title/caption, date and source context.
- Adding from a class record auto-fills class/date/subject context.
- One Moment can link to several students without duplicating the media.
- Moments contains photos/videos, not development journaling.

Official desktop + mobile UI reference



Flow 12 — Moments

Source-of-truth rule for this flow: the written specification controls behavior, permissions and data meaning; this board controls intended visual structure, responsive layout and screen states. If generated sample names/dates/copy conflict with a locked MVP rule, the written rule takes precedence.

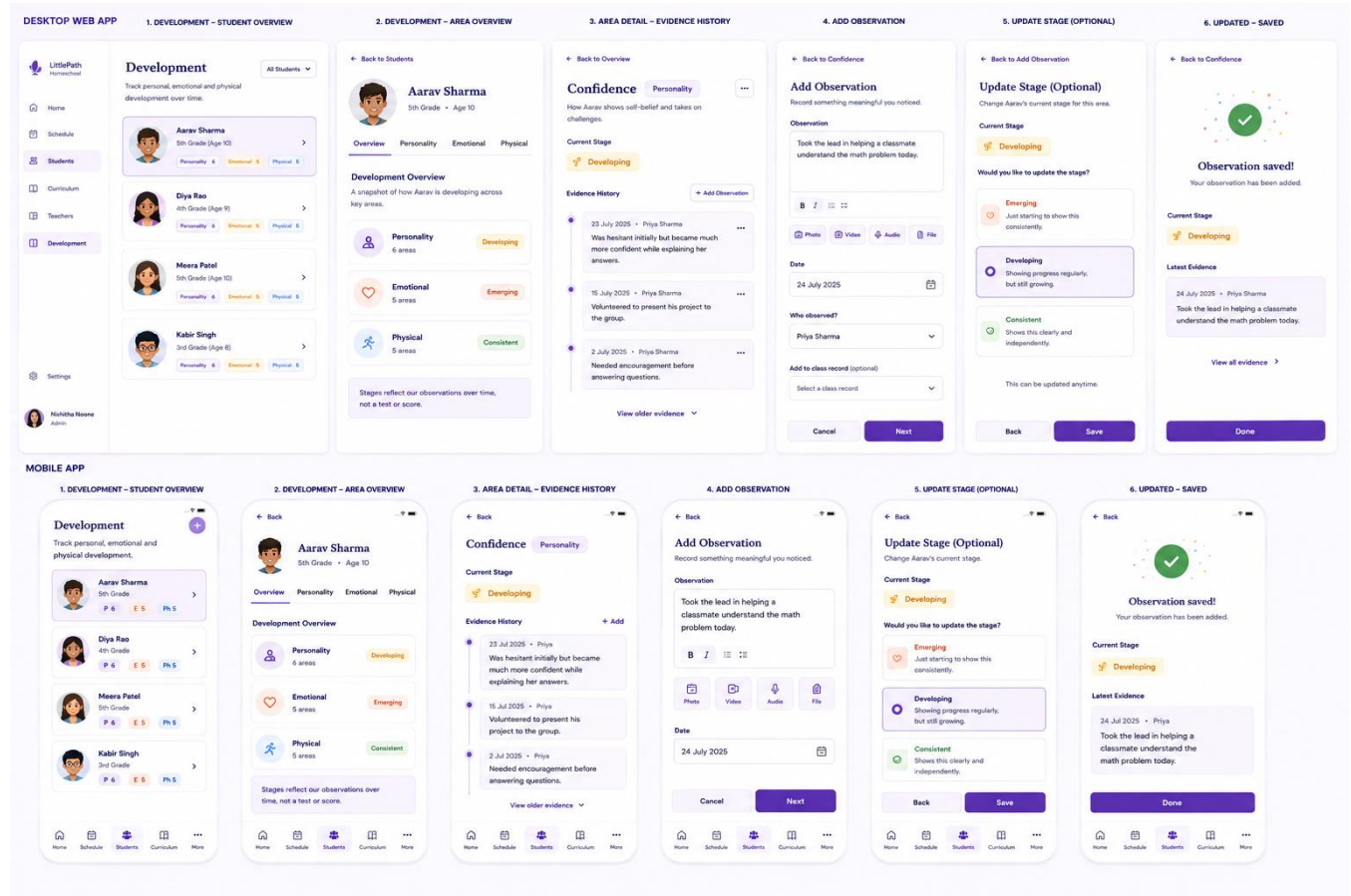
Flow 13 — Development

Primary role	Admin / Teacher
Purpose	Evidence-based development observations.

Locked behavior

- Development is separate from academic Learning and Moments.
- Group areas under Personality, Emotional and Physical.
- Each area has evidence history and a stage such as Emerging, Developing or Consistent.
- Observation may link evidence/source class; stage update is optional and remains human judgment, not automatic AI scoring.

Official desktop + mobile UI reference



Flow 13 — Development

Source-of-truth rule for this flow: the written specification controls behavior, permissions and data meaning; this board controls intended visual structure, responsive layout and screen states. If generated sample names/dates/copy conflict with a locked MVP rule, the written rule takes precedence.

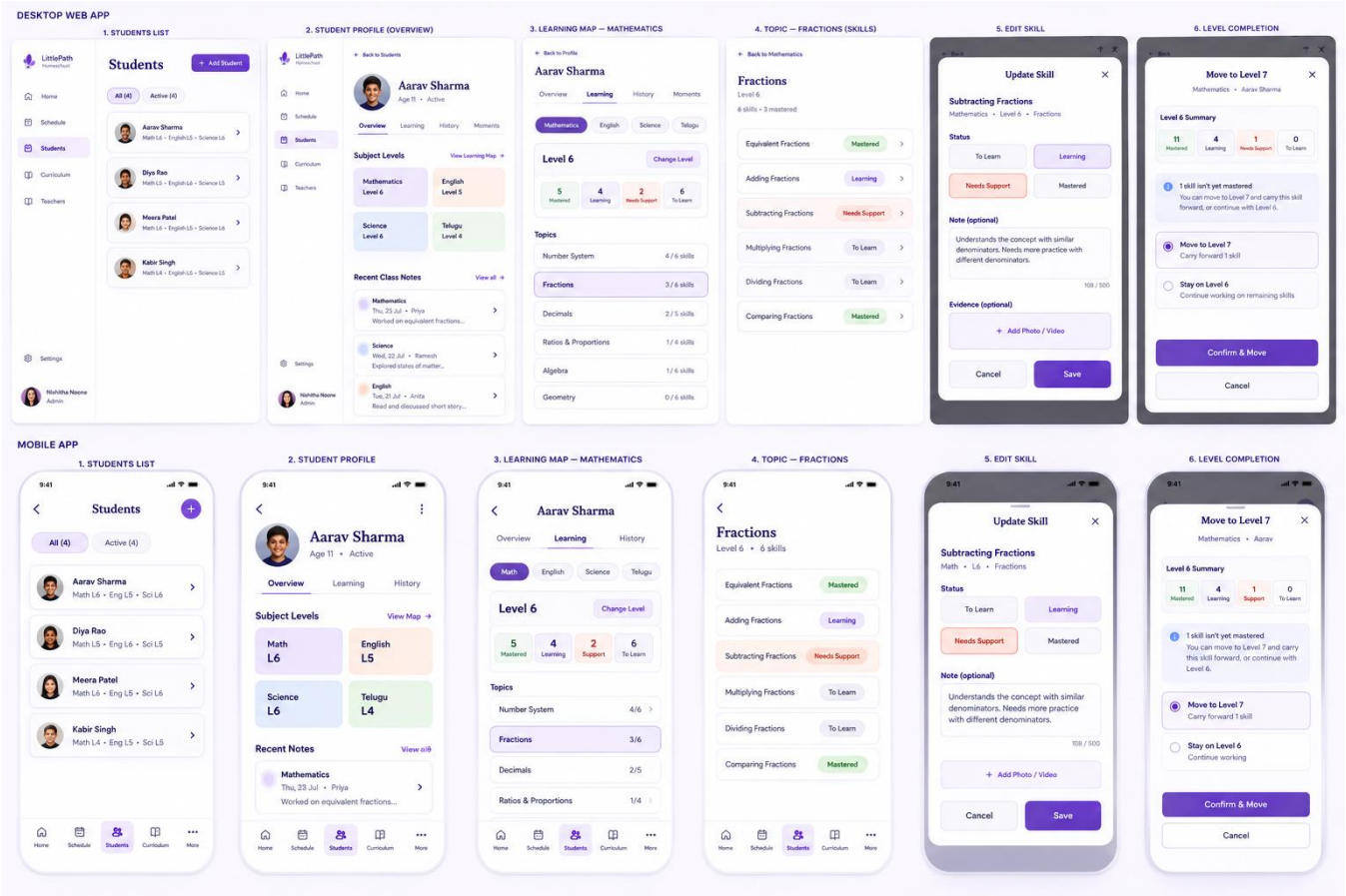
Flow 14 — Student Learning Map & Skill Progress

Primary role	Admin / Teacher
Purpose	Track academic curriculum progress per student.

Locked behavior

- Student-specific statuses: To Learn, Learning, Needs Support, Mastered.
- Authorized staff can edit status and optional note/evidence.
- Level completion warns about unfinished skills and previews carry-forward impact.
- Curriculum defines what exists; Learning Map defines where the student is.

Official desktop + mobile UI reference



Flow 14 — Student Learning Map & Skill Progress

Source-of-truth rule for this flow: the written specification controls behavior, permissions and data meaning; this board controls intended visual structure, responsive layout and screen states. If generated sample names/dates/copy conflict with a locked MVP rule, the written rule takes precedence.

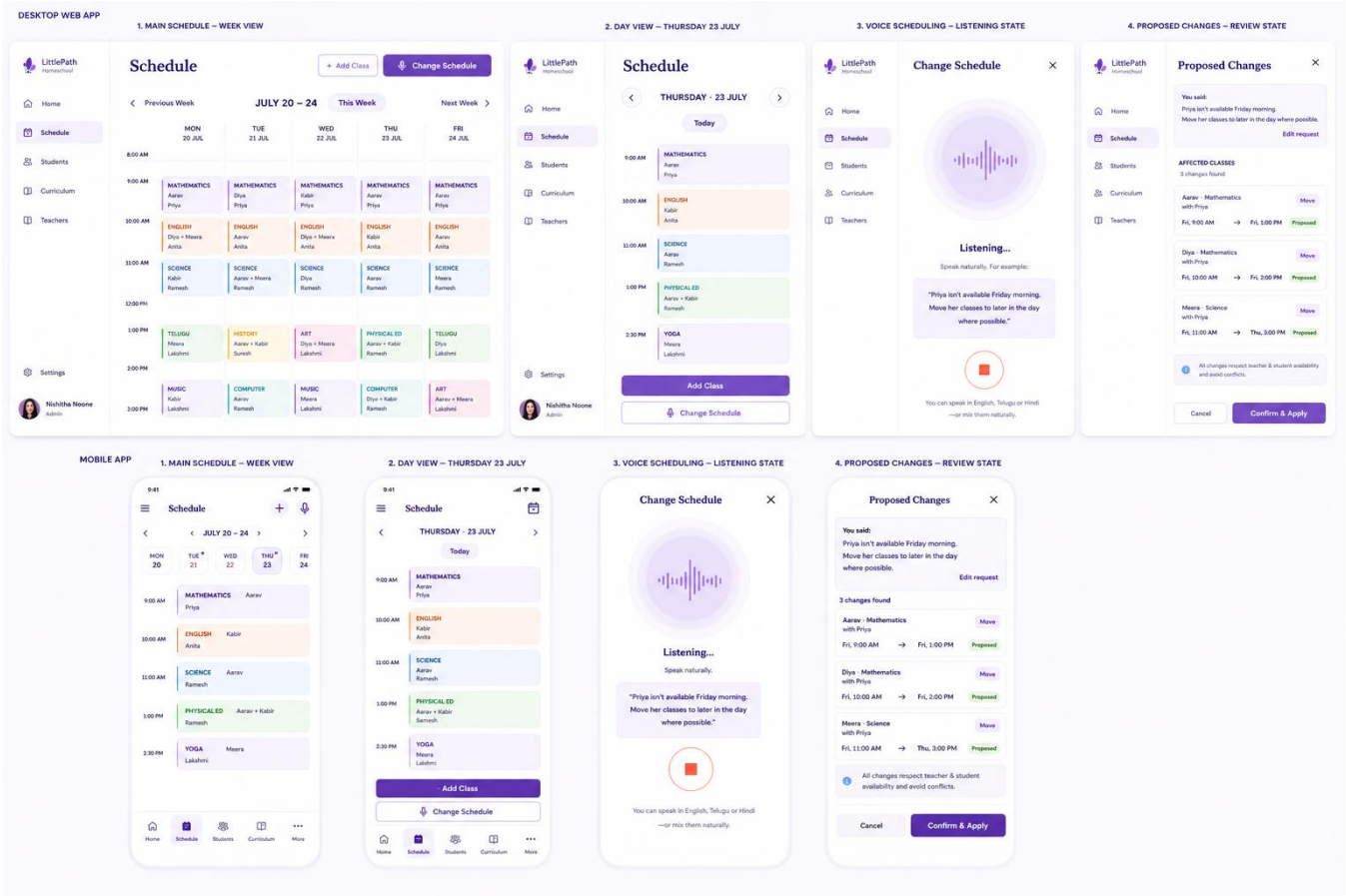
Flow 15 — Schedule Management & Voice Rescheduling

Primary role	Admin
Purpose	View schedule and propose multi-class changes.

Locked behavior

- Week/day schedule plus Add Class and Change Schedule.
- Voice request can express teacher unavailability or multi-class movement.
- Show every proposed move and affected class before applying.
- Revalidate teacher/student availability and conflicts; admin confirms before mutation.

Official desktop + mobile UI reference



Flow 15 — Schedule Management & Voice Rescheduling

Source-of-truth rule for this flow: the written specification controls behavior, permissions and data meaning; this board controls intended visual structure, responsive layout and screen states. If generated sample names/dates/copy conflict with a locked MVP rule, the written rule takes precedence.

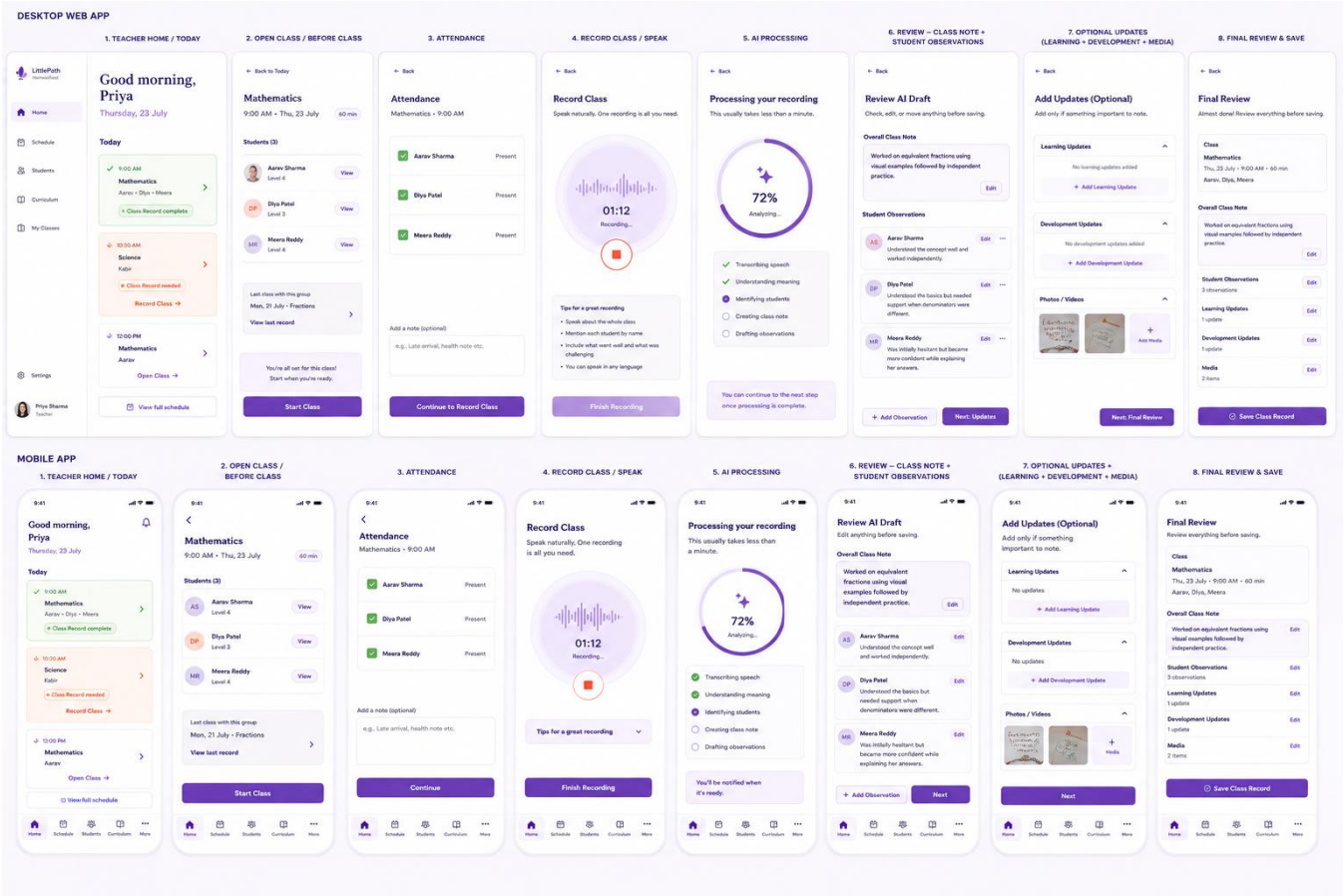
Flow 16 — Teacher Post-Class Workflow

Primary role	Teacher
Purpose	Teacher-first end-to-end class documentation flow.

Locked behavior

- Teacher Home → Open Class → Attendance → Record/Speak → AI Processing → Review Draft → Optional Updates → Final Review & Save.
- Before recording, show group and previous class record for continuity.
- Attendance is embedded in this flow.
- One recording covers whole class and named children; AI splits Overall Class Note and Student Observations.
- Learning/Development/media updates are optional; never force teachers to manufacture updates.
- Final save preserves Overall Class Note regardless of optional updates.

Official desktop + mobile UI reference



Flow 16 — Teacher Post-Class Workflow

Source-of-truth rule for this flow: the written specification controls behavior, permissions and data meaning; this board controls intended visual structure, responsive layout and screen states. If generated sample names/dates/copy conflict with a locked MVP rule, the written rule takes precedence.

7. Cross-Flow Business Rules

- Historical class occurrences/records remain stable when future schedules, levels or availability change.
- Availability is a scheduling constraint; attendance records what actually happened.
- Schedule validity requires teacher capability, teacher availability and student availability, with no unresolved conflicts.
- Learning and Development changes preserve history, author/date/source and are not destructive overwrites.
- A Moment tagged to multiple students is one media object with multiple links.
- Parent data is scoped to linked children; teacher access is scoped to relevant teaching data; admin manages the school workspace.

8. Notification Rules

Parent	One weekly notification only: weekly update ready.
Teacher	Relevant scheduling changes/issues and class-record/class-update reminders.
Admin	Actionable operational issues such as conflicts, availability changes and incomplete setup.

9. Non-Functional MVP Requirements

- Responsive desktop and phone implementation matching the boards.
- Server-side role authorization.
- Modern password hashing and forced replacement of temporary password.
- Object storage for media; database stores references and metadata.
- AI output remains untrusted draft data until confirmed.
- Preserve author/source/timestamps for learning, development and class-record history.
- Optimize for the current small-school scale; avoid premature enterprise complexity.

10. Implementation Order

1. Authentication + authorization
2. Curriculum
3. Teacher management + availability
4. Student management + parent links + availability
5. Scheduling + constraint engine
6. Home states
7. Class occurrences + attendance
8. Voice transcription + AI extraction
9. Class record review/save
10. Learning Map
11. Development
12. Moments/media
13. Parent portal + weekly update
14. Notifications
15. System states

11. MVP Acceptance Checklist

- ☐ Admin can complete setup and schedule classes.
- ☐ AI-assisted scheduling returns valid options and never applies changes without confirmation.

- ☐ Teacher can mark attendance and create a class record from one voice recording.
- ☐ Voice processing creates both an overall Class Note and separated student observations.
- ☐ Overall Class Note saves even with no optional student updates.
- ☐ Learning, Development and Moments remain distinct.
- ☐ Parent sees only linked-child Home/Learning/Development/Moments/weekly update, with no Schedule or Ask Teacher.
- ☐ Parent gets one weekly notification; teacher notifications stay limited to scheduling and class-record needs.
- ☐ Login uses username/password without OTP and forces first-login password change.
- ☐ Empty/loading/error/offline states are implemented.
- ☐ Desktop/mobile UI follows the 16 embedded boards.

12. Deferred / V2

- Parent schedule access
- Ask Teacher / messaging
- Library integration
- Global search
- Advanced reporting/analytics
- Homework/assignments
- Additional integrations
- Broader AI recommendations/automation

13. Final Source of Truth

This document is the MVP source of truth. Written rules control product behavior, permissions, data meaning and scope. Embedded UI boards control the intended visual implementation. Future changes should be documented explicitly rather than inferred from mockups.